

Multiple Choice

1. **Answer:** D
Options: A) 0.1; B) 0.2; C) 0.3; D) 0.5
Note: Correct option: D - 0.5
2. **Answer:** C
Options: A) Abstract Syntax Tree (AST); B) Attribute Grammar; C) Symbol Table; D) Semantic Stack
Note: Correct option: C - Symbol Table
3. **Answer:** B
Options: A) 0; B) 1; C) 3; D) 4
Note: Correct option: B - 1
4. **Answer:** A
Options: A) Authentication; B) Confidentiality; C) Integrity; D) Nonrepudiation
Note: Correct option: A - Authentication
5. **Answer:** A
Options: A) isupper(); B) join(seq); C) len(string); D) ljust(width[, fillchar])
Note: Correct option: A - isupper()

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'All bit strings with more ones than zeros'.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Replace the page whose next reference will be the longest time in the future.'

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Merge sort'.

Long Form Response

11. **Answer:** `def mutiple_tuple(nums): | return product`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def replace_char(str 1,ch,newch): | return str 2`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key